

CRDs with Multiple Arms

PUBH 526

Design and Analysis of Randomized Trials in Public Health

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CRD with a Treatments

- Suppose you have a treatments/study arms

$$y_{ij} = \mu_i + \varepsilon_{ij} \begin{cases} i = 1, 2, \dots, a \\ j = 1, 2, \dots, n_i \end{cases}$$

$$\varepsilon_{ij} \sim iid N(0, \sigma^2)$$

- $\mu_i = i^{\text{th}}$ treatment mean
- $H_0: \mu_1 = \mu_2 = \dots = \mu_a$
- $H_A: \text{at least one of the } \mu_i \text{ is different}$

Alternative Parameterization

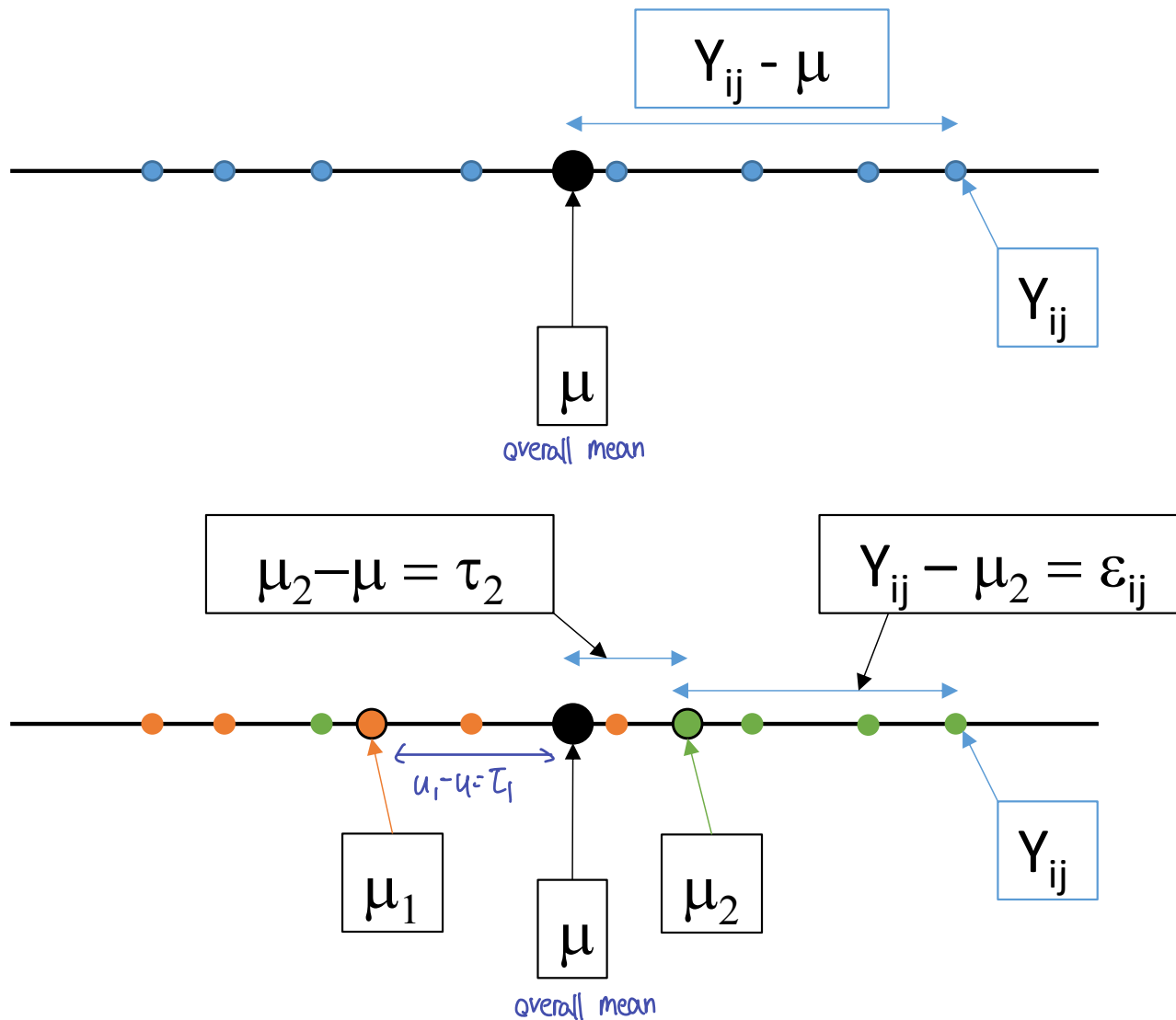
- Suppose you have a treatments/study arms

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij} \begin{cases} i = 1, 2, \dots, a \\ j = 1, 2, \dots, n_i \end{cases}$$

$$\varepsilon_{ij} \sim iid N(0, \sigma^2)$$

- μ = overall mean
- $\mu + \tau_i = i^{\text{th}}$ treatment mean
- τ_i = effect of treatment i (how the i^{th} treatment mean differs from the overall mean)
- $H_0: \tau_1 = \tau_2 = \dots = \tau_a = 0$ All same as μ .
- $H_A: \tau_i \neq 0$ for at least one i

Why? – Partitioning the Variability of Y_{ij}



Partitioning the Variability of Y_{ij}

- Notation:

- $\bar{y}_{i.}$ = sample mean for treatment i
- $\bar{y}_{..}$ = overall mean of your sample

- $y_{ij} = \mu + \tau_i + \varepsilon_{ij}$

- $y_{ij} - \mu = \tau_i + \varepsilon_{ij}$

- $y_{ij} - \bar{y}_{..} = (\bar{y}_{i.} - \bar{y}_{..}) + (y_{ij} - \bar{y}_{i.})$

Partitioning the Variability of Y_{ij}

- Total sums of squares = treatment sums of squares + error sums of squares

$$\sum \sum (y_{ij} - \bar{y}_{..})^2 = \sum n_i (\bar{y}_{i.} - \bar{y}_{..})^2 + \sum \sum (y_{ij} - \bar{y}_{i.})^2$$

$$SS_T = SS_{\text{Treatments}} + SS_E$$

- To test H_0 , look at

- If H_0 is true, all those $(\bar{y}_{i.} - \bar{y}_{..})$ will be close to zero
($SS_{\text{treatments}}$ close to zero)
- If $SS_{\text{treatments}}$ is large, we'll want to reject H_0
- How large is "large"?

◦ meaning $\bar{y}_i \approx \bar{y}$, so no difference, fail to reject H_0

F Statistic!

- Standardize those sums of squares:

$$\begin{aligned}
 F_0 &= \frac{\sum n_i (\overset{\text{SS Regression}}{\bar{y}_{i.}} - \bar{y}_{..})^2 / (a - 1)}{\sum \sum (y_{ij} - \bar{y}_{i.})^2 / (N - a)} \\
 &= \frac{\text{SS}_{\text{Treatment}} / (\overset{\substack{\text{\# of parameters} \\ p-1}}{a - 1})}{\underset{n-p}{\text{SS}_E / (N - a)}} = \frac{MS_{\text{Trt}}}{MS_{\text{Error}}} \\
 &= \frac{\text{“average” squared deviation between trts}}{\text{“average” squared deviation within trts}}
 \end{aligned}$$

Why use F_0 ?

- From general sum of squares result, can show
 1. $SS_E/(N - a)$ is unbiased estimate of σ^2
 2. Under H_0 , $SS_{\text{Treatment}}/(a - 1)$ is also unbiased estimate of σ^2

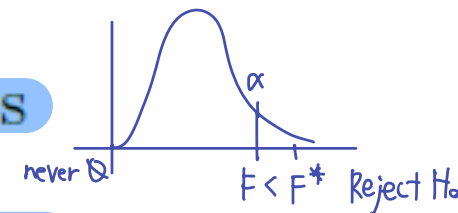
- Call these estimates **Mean Squares**

- Can show the expected values are

$$E(MS_E) = \sigma^2$$

$$E(MS_{\text{Treatment}}) = \sigma^2 + \sum n_i \tau_i^2 / (a - 1)$$

- The test statistic **F_0 is a ratio of mean squares**



- If statistic much **larger than one, then reject H_0**

- What is the test statistic's distribution under H_0 ?

Analysis of Variance Table

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
Between	$SS_{\text{Treatment}}$	$a - 1$	$MS_{\text{Treatment}}$	F_0
Within	SS_E	$N - a$	MS_E	
Total	SS_T	$N - 1$		

If $F_0 > F_{\alpha, a-1, N-a}^{df_1, df_2}$ then reject H_0 (using critical value)

If $P(F_{a-1, N-a} > F_0) \leq \alpha$ then reject H_0 (using P -value)

anova2.sas

- What would it mean if we left out the class statement in proc glm?

Hypotheses About Specific Means

$$y_{ij} = \mu_i + \varepsilon_{ij} \begin{cases} i = 1, 2, \dots, a \\ j = 1, 2, \dots, n_i \end{cases}$$

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij} \begin{cases} i = 1, 2, \dots, a \\ j = 1, 2, \dots, n_i \end{cases}$$

- $H_0: \mu_1 = \mu_2 = \dots = \mu_a$
- $H_0: \tau_1 = \tau_2 = \dots = \tau_a = 0$
- However, we're usually interested in more specific hypotheses

Hypothesis Testing depends on how we specify the model (u v.s. T)

Recall:

- Regression approach for 2-arm study:

$$y_j = \beta_0 + \beta_1 X + \varepsilon_j$$

$$\varepsilon_j \sim iid N(0, \sigma^2)$$

- $j = 1, 2, \dots, \Sigma n_i$ (full sample, across all groups)
- $X=1$ if observation is in the treatment group,
0 otherwise
- $H_0: \beta_1=0$
- β_0 =mean in the control group

How Would We Do This With 3+ Arms?

Indicator variables: (Arms-1)

- $X_1=1$ if observation is in the treatment group 1,
0 otherwise
- $X_2=1$ if observation is in the treatment group 2,
0 otherwise
- ... $X_3=0,0$
- How many such indicator variables do we need to create?

2 for 3 Arms

$$\begin{aligned} X_1 &= (1, 0) \\ X_2 &= (0, 1) \\ X_3 &= (0, 0) \end{aligned}$$

$$\hat{Y} = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$$

$$X_{1j} = 1 \text{ if trt 1} \\ \emptyset \text{ otherwise}$$

$$X_{2j} = 1 \text{ if trt 2} \\ \emptyset \text{ otherwise}$$

“anova2.sas”

$$\hat{Y} = 141.6 - 11.2X_1 - 8.6X_2$$

Interpretation of β 's:

$X_1 = (1, 0)$	$\hat{Y} = 141.6 - 11.2 = 130.4 \rightarrow 11.2$ lower than Baseline (Trt 3)
$X_2 = (0, 1)$	$\hat{Y} = 141.6 - 8.6 = 133.0 \rightarrow 8.6$ lower than Baseline (Trt 3)
$X_3 = (0, 0)$	$\hat{Y} = 141.6$

- This is actually what PROC GLM does

```
title1 'PROC GLM with "solution" statement';  
proc glm data=anova1;  
    class treatment;  
    model endline_glucose = treatment / solution;  
run;  
title1;
```

```
title1 'PROC GLM specifying treatment 1 as the control/comparison group';  
proc glm data=anova1;  
    class treatment(ref="1");  
    model endline_glucose = treatment / solution;  
run;  
title1;
```

Assumptions

1. Independent observations – errors are independent
2. Errors are normally distributed
3. Errors have constant variance

➤ Residuals are estimates of unexplained noise

Errors Are Independent

- Plot residuals vs. time and/or space
 - Is there a drift or pattern as time goes on?
- Plot residuals vs. relevant variables
 - Variables omitted from your model
 - May identify new factors for secondary analyses or future studies

Normally Distributed Errors

- Histogram of residuals
 - Symmetric, bell shaped?
- Normal probability plot / QQ plot
 - Does the relationship look approximately linear?
- Shapiro-Wilks / Kolmogorov-Smirnov test

Errors Have Constant Variance

- Be aware: Variance of a response is often linked to its mean
- Plot residual vs. predicted values
 - Is the range of the residuals constant across the predicted values?
- Modified Levene's test (Brown and Forsythe's test)

Outliers

- Investigate!
 - Data entry error?
 - Error in measurement?
 - Actual possible value
 - Consistency with other variables
 - Something new and interesting?
- Omitting means leaving out an entire observation – don't do this easily
- Are the outliers influential?
- Do analysis with and without outlier(s)
 - Do your conclusions change?
 - How much do your estimates change?
- Standardized (studentized) residuals help identify unusual values but offer no explanations

Constant Variance

- Non-constant variance happens
- F test is robust to violations, especially if design is balanced (same sample size in each arm)
- However, comparison of treatments depends on MSE
 - Non-constant variance affects your estimate of MSE, in turn affecting your hypothesis tests and confidence intervals
- Box-Cox transformation: Y^λ
 - $\ln(Y)$ when $\lambda=0$
 - Stabilizes variance, often improves normality as well

boxcox.sas

- This may or may not solve the problem
 - Check diagnostics again after transformation
 - May need to consider other transformations or models

Box-Cox Transformations

- λ can be any real number
- SAS suggests a value for λ but also a CI for λ
 - A nearby value for λ may make more “sense”
- Handling zeroes
 - $\ln(x+0.001)$

Box-Cox Transformations

- People have a hard time thinking in terms of transformed variables ($\sqrt{\text{blood glucose}}$?)
- Means of transformed variables, and changes in those means, are hard to interpret/act upon
- Not easy to convert back to original scale: $\text{Mean}(\sqrt{Y}) \neq \sqrt{\text{Mean}(Y)}$
- Suggestion:
 - Summary statistics and plots on untransformed variable
 - P-values on transformed variable
 - Discussion/interpretation on untransformed variable

Nonparametric One-Way ANOVA

- Kruskal-Wallis test – when normality assumption may not be satisfied
- H_0 : All of the a population distribution functions are identical
- H_A : At least one of the population distributions generates values higher than the others
- See “kruskal-wallis.sas”

Aside ...

- Suppose you have a randomized study with:
 - No differences at baseline for your primary outcomes, key potential confounders or effect modifiers
 - Very little loss to follow-up or missing data
 - Adequate sample size based on your initial power calculations
 - Statistically significant differences in your primary outcome after the interventions
- Do you automatically attribute the differences in outcome to the interventions?

Aside ...

- Do you automatically attribute the differences in outcome to the interventions?
 - Context (example: malaria outbreak in part of study area)
 - Mechanism (examples: behavior change intervention but no attendance, change in functional outcome but no change in biomarker)
 - Role of delivery mechanism (example: placebo effect from group meetings)
 - Multiple messages (example: non-target behavior changed, and that led to improved outcome)